

Finding a Signal

From a table of collisions to a discovery — Chapter 8

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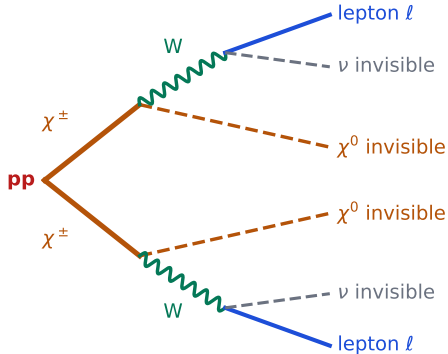
Part 1 — The SUSY dataset

What you're getting

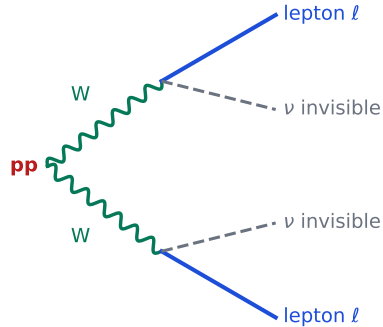
- ▶ **5 million simulated collisions**, one per row, in a 2.2 GB CSV file
- ▶ Half are a hypothetical **supersymmetry** signal, half a well-known **background**
- ▶ Each row says which one it really was: the **truth**
- ▶ From the first paper to show that **deep learning** could do physicists' work: Baldi, Sadowski & Whiteson, *Searching for exotic particles in high-energy physics with deep learning* (2014)

What was simulated

signal: a pair of charginos



background: a pair of W bosons



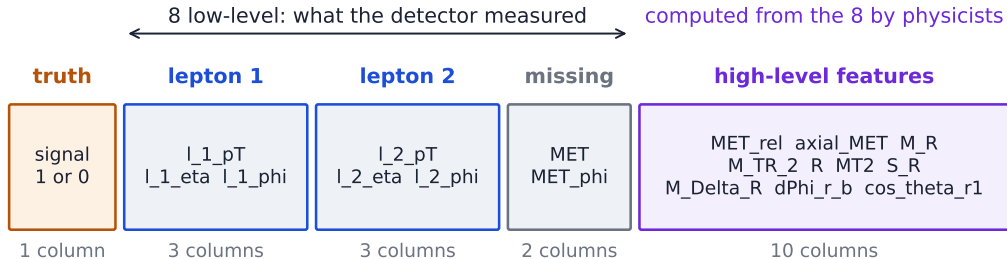
both leave two leptons + missing momentum; the signal has two extra, heavy, invisible particles

- Same visible particles in both. The difference is **how they move**, not what they are

From a picture to a row

- ▶ The detector takes a picture; **reconstruction** turns it into a list of particles
 - ▶ electrons, muons, photons, jets, each with a measured momentum
 - ▶ plus one event-level number: the **missing momentum**
- ▶ Keep only collisions with **exactly two leptons** (electrons or muons)
- ▶ Write one row: who was there, how they moved
- ▶ A **lepton** is an electron or a muon; the file doesn't say which

One row per collision



- **19 numbers per row.** The file has **no header**: the column names come from the paper (and Lab 7)

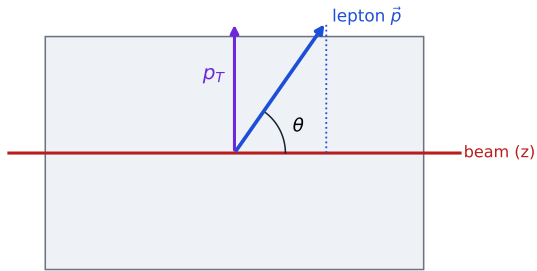
The first five rows

signal	l_1_pT	l_1_eta	l_1_phi	l_2_pT	l_2_eta	l_2_phi	MET	MET_phi
0	0.973	0.654	1.176	1.157	-1.740	-0.874	0.568	-0.175
1	1.668	0.064	-1.225	0.506	-0.339	1.673	3.475	-1.219
1	0.445	-0.134	-0.710	0.452	-1.614	-0.769	1.220	0.504
1	0.381	-0.976	0.693	0.449	0.892	-0.677	2.033	1.533
1	1.310	-0.690	-0.676	1.589	-0.693	0.623	1.088	-0.382

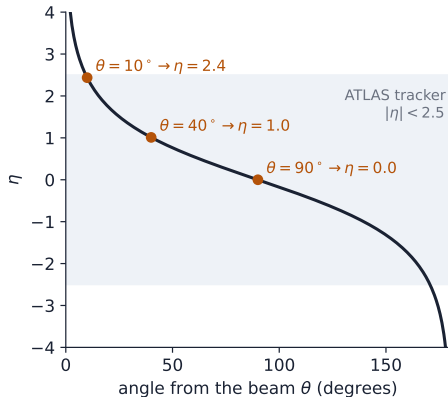
... and 10 high-level columns to the right; 5,000,000 rows, no header

- Signal and background are **mixed**, in no order. Column 0 tells you which is which

A funny coordinate system: p_T , η , ϕ



p_T = momentum across the beam; $\eta = -\ln \tan(\theta/2)$;
 ϕ = angle around the beam (in and out of the page)



- Same information as (p_x, p_y, p_z) : **three numbers** for a momentum, just different ones

Why *transverse*?

- ▶ Protons are bags of quarks: the two colliding pieces carry **unknown** shares of the beam energy
- ▶ So momentum **along the beam** isn't known before the collision...
- ▶ ...but momentum **across the beam** starts at **zero** — and must still add up to zero after
- ▶ p_T is what you can balance. η differences don't change when the whole event slides along the beam

Missing momentum: MET, MET_phi

- ▶ Add up the transverse momenta of everything seen. It should be zero
- ▶ Whatever is missing went into **invisible** particles: neutrinos, and (in the signal) neutralinos
- ▶ MET = how much is missing; MET_phi = which direction
- ▶ Only **two** numbers: along the beam, nothing can be balanced

The units: everything is rescaled

- ▶ The values are **not** in GeV or radians. In this file:
 - ▶ every momentum-like column has a mean of about **1.0**
 - ▶ `l_1_phi` runs from **-1.73 to $+1.73$** , not $-\pi$ to $+\pi$
- ▶ Shapes, orderings and signal-vs-background differences are intact
- ▶ Don't read a value as a physical quantity: compare distributions

The truth column

- ▶ `signal = 1` for supersymmetry, 0 for background
- ▶ **46% signal** in the file. Among real collisions, a signal like this is 1 in 10^{11} or rarer
 - ▶ balanced on purpose, to train and test classifiers
- ▶ The truth is the **answer key**: use it to measure how well a method does, **never** as an input to the method

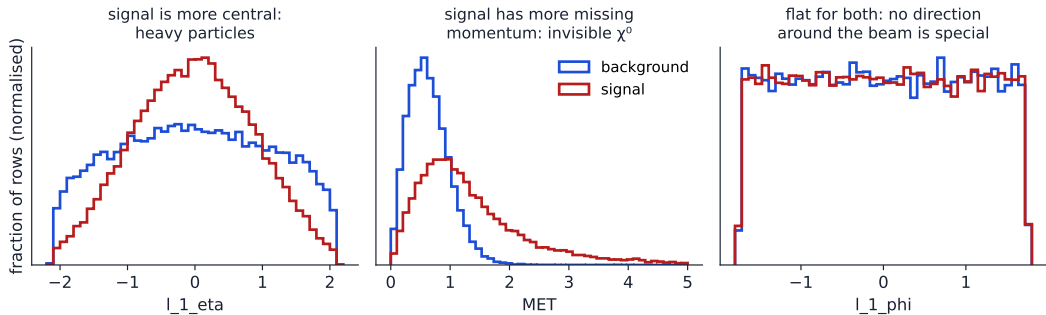
High-level features

- ▶ **Low-level:** the 8 numbers the detector measured
- ▶ **High-level:** quantities physicists **computed** from those 8, designed so signal and background differ
- ▶ Example: add two leptons' four-vectors and get a mass near 91 GeV $\rightarrow$ probably a **Z boson**
 - ▶ if Z's are your background, reject those collisions: a "Z veto"
- ▶ Careers were spent inventing these. That's **feature engineering**

The ten high-level features, roughly

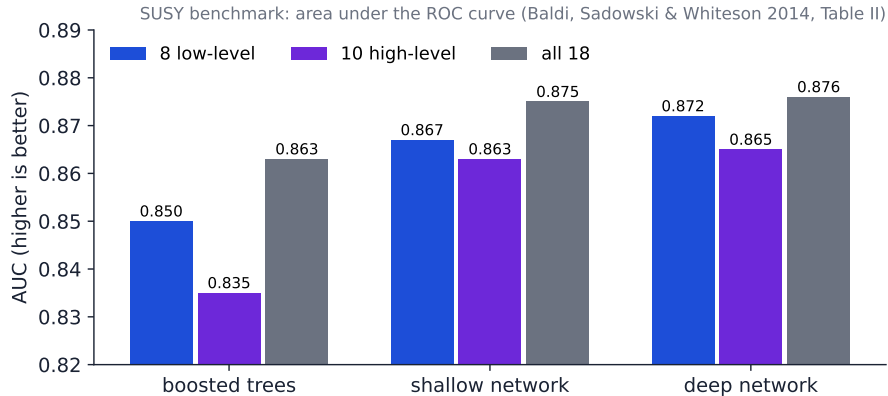
- ▶ **How much is missing, refined:** MET_{rel} (reduced when it points close to a lepton or jet, where it may be mismeasured), axial_MET (missing momentum along the leptons' axis)
- ▶ **Mass-scale estimates** of the heavy pair: M_R , M_{TR_2} and their ratio R ("razor"); MT2 ("stransverse mass"); S_R and $M_{\Delta R}$ ("super-razor")
- ▶ **Angles** in the frame of the heavy pair: $d\Phi_{r_b}$, $\cos_{\theta_{r1}}$
- ▶ You don't need the formulas. You need to know they are **functions of the other eight**

What tells them apart



- Domain knowledge predicts all three. First 50,000 rows of the real file

Did the hand-made features help?



- ▶ A **deep network on the 8 raw numbers** beat simpler methods on the 10 hand-made ones

In Lab 7

- ▶ **Download once**, then comment the download cell out (2.2 GB)
- ▶ Shrink it: `head -500000 SUSY.csv > SUSY-small.csv` — and use the small file
- ▶ Give the columns names; split rows into signal and background with the truth column
- ▶ Look before you compute: histograms of every column, signal vs background, **normalised**
- ▶ Then: correlations, then choosing **cuts** (Lecture 16)

Recap of Part 1

- ▶ One row per collision: truth, two leptons (p_T, η, ϕ), missing momentum, 10 computed features
- ▶ Signal and background look alike; they differ in **how** things move
- ▶ Values are rescaled; the truth column is an answer key, not an input
- ▶ Hand-made features vs learned ones: the question the paper answers

Next: counting, cuts and significance (Lecture 16)